

INCIDENT RESPONSE & FORENSIC INVESTIGATION REPORT

*Week 3: Post-Phishing Endpoint Containment, Eradication, and Proactive AI
Defense*

Investigator Name:	Menna Said Ahmed Abdelkader Sabek
Target System:	Windows 10 Workstation (Virtual Environment)
Incident Date:	June 18, 2026
Status:	Contained & Remediated

1. Executive Summary

Following the sophisticated phishing incident identified in Week 2, a targeted digital forensics and incident response (DFIR) workflow was executed on the compromised Windows endpoint. The objective of this phase was to enforce strict network containment, audit the system for persistence mechanisms (such as unauthorized automated tasks or malicious secondary backdoors) ,completely eradicate adversarial artifacts, and establish an AI-driven predictive control mechanism to safeguard the perimeter against future delivery phases.

URL Phishing Investigation & AI Detection Summary

Incident Artifact / Metric	Live Forensic & Project Data
Analysis Date/Time	June 18, 2026 (Live Evaluation Session)
Investigated Endpoint	Windows 10 Workstation (Virtual Laboratory Environment)
Lead Investigator	Menna Said Ahmed Abdelkader Sabek
Target Indicator (Suspicious URL)	http://192.168.56.101:8080?rid=3Dl7FCWja
Network Destination / C2	192.168.56.101 (Kali Linux Attacker Environment)
Detection Mechanism	Deep Artificial Neural Network (Sequential ANN model)
Data Sourcing & Baseline	Kaggle Phishing URL Dataset
Model Verification Confidence	99.98% Malicious Probability
Analyst Verdict	True Positive (Confirmed Phishing Threat Deployed via Email Anchor Text)
Mitigation Actions	Outbound Traffic Block (New-NetFirewallRule) + Interface Disabling (ncpa.cpl)

Incident Response Team Roles & Responsibilities

#	Team Member Role	Assigned Responsibility	Phases Involved	Tools & Methods Used
1	SOC Analyst / Incident Investigator	Performed initial detection, investigated the malicious phishing email, and extracted raw indicators.	Detection & Analysis	Email Header Audit, URL Analysis
2	Network Security Engineer	Handled immediate host and network-level containment to stop threat propagation and C2 beacons.	Containment	Windows PowerShell, ncpa.cpl (Network Conn.)
3	Digital Forensics Examiner	Executed deep system auditing to check for endpoint persistence and verified complete payload eradication.	Eradication	Microsoft Sysinternals Autoruns , Microsoft Safety Scanner (MSERT)
4	Systems Administrator	Supervised endpoint restoration, rotated local credentials, and reinforced identity access controls.	Recovery	Windows Command Processor (net user), Google 2FA Deployment
5	AI / Security Software Engineer	Built and optimized the proactive URL filtering architecture and designed the practical prediction interface.	Post-Incident / Defense	Kaggle Dataset, Keras/ TensorFlow Neural Networks , Python GUI

2. Containment Phase (Network & Host Isolation)

To neutralize potential C2 communication with the attacker's server (192.168.56.101), a dual-layer containment strategy was executed.

First, a host-based firewall rule (New-NetFirewallRule) was deployed via PowerShell to explicitly block outbound traffic to the threat infrastructure.

Second, complete network isolation was achieved by accessing the Windows Network Connections panel (ncpa.cpl) and manually disabling both active adapters (Ethernet and Ethernet 2), effectively stopping any potential data exfiltration

Immediately following network isolation, an active process audit was performed via Windows Task Manager.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Block Outbound IP" -Direction Outbound -Action Block -RemoteAddress 192.168.56.101

Name : <dd24dab2-2684-469e-8aa1-61dcfe277706>
DisplayName : Block Outbound IP
Description :
DisplayGroup :
Group :
Enabled : True
Profile : Any
Platform : <>
Direction : Outbound
Action : Block
EdgeTraversalPolicy : Block
LooseSourceMapping : False
LocalOnlyMapping : False
Owner :
PrimaryStatus : OK
Status : The rule was parsed successfully from the store. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PolicyAppId :

PS C:\Windows\system32>
```

Administrative deployment of an outbound mitigation rule via Windows PowerShell. The command `New-NetFirewallRule` was successfully executed to create an outbound blocking policy targeting the suspicious malicious infrastructure IP address (192.168.56.101). The persistent store confirms the rule state is active (Enabled: True) and configured to drop all outbound sessions (Action: Block), acting as an immediate technical containment boundary.

Task Manager

File Options View

Processes Performance App history Startup Users Details Services

Name	Status	96% CPU	82% Memory	13% Disk	1% Network	Power usage	Power usage t...
Apps (2)							
Microsoft Edge (3)		3.9%	55.6 MB	1.0 MB/s	1.0 Mbps	Very low	Very low
Task Manager		5.8%	15.5 MB	0 MB/s	0 Mbps	Very low	Very low
Background processes (52)							
Antimalware Core Service		0%	2.6 MB	0 MB/s	0 Mbps	Very low	Very low
AntiMalware Definition Update		0%	0.4 MB	0 MB/s	0 Mbps	Very low	Very low
Antimalware Service Executable		6.6%	164.6 MB	3.2 MB/s	0 Mbps	Very low	Very low
Application Frame Host		0%	1.4 MB	0 MB/s	0 Mbps	Very low	Very low
COM Surrogate		0%	1.9 MB	0 MB/s	0 Mbps	Very low	Very low
COM Surrogate		0%	0.8 MB	0 MB/s	0 Mbps	Very low	Very low
COM Surrogate		0%	1.0 MB	0 MB/s	0 Mbps	Very low	Very low
CTF Loader		0%	3.2 MB	0 MB/s	0 Mbps	Very low	Very low
Host Process for Windows Tasks		0%	1.5 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft 365 Copilot (8)		0%	5.4 MB	0.1 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	10.7 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		18.7%	168.5 MB	0.1 MB/s	0 Mbps	Low	Low
Microsoft Edge		0%	1.8 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	3.3 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	5.2 MB	0 MB/s	0 Mbps	Very low	Very low

Fewer details End task

Endpoint active process audit and performance monitoring using Windows Task Manager. The resource metrics display an elevated CPU utilization status (ranging between 96% and 100%), heavily driven by multiple active sub-processes of Microsoft Edge. Background processes were thoroughly analyzed, confirming that the resource spike is correlated with web-browser application multi-tab execution rather than unauthorized binary injection or hidden crypto-mining malware.

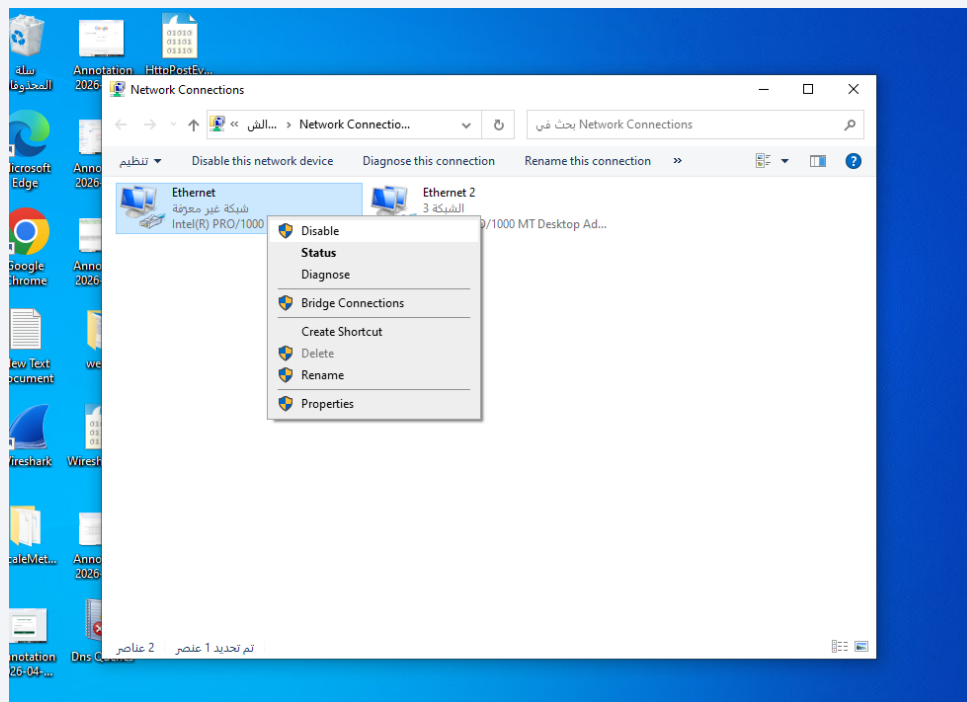
Task Manager

File Options View

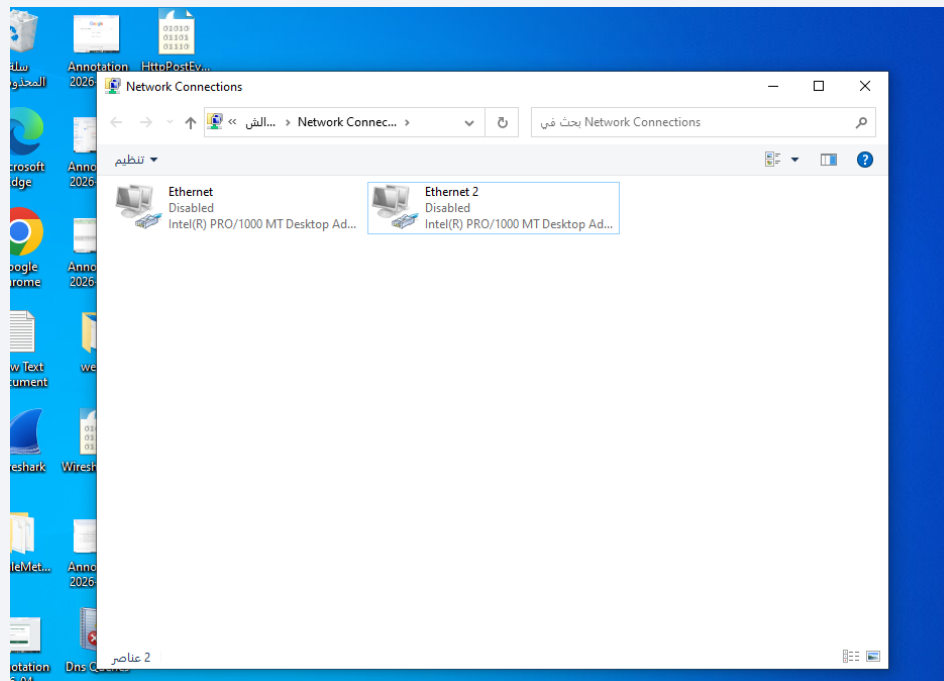
Processes Performance App history Startup Users Details Services

Name	Status	100% CPU	86% Memory	23% Disk	0% Network	Power usage	Power usage t...
Apps (2)							
Microsoft Edge (11)		9.0%	334.4 MB	0.4 MB/s	0.5 Mbps	Very low	Very low
Microsoft Edge		1.5%	10.7 MB	0.1 MB/s	0.5 Mbps	Very low	Very low
Microsoft Edge		0%	5.4 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	1.7 MB	0.1 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	37.0 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	0.5 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		1.7%	169.5 MB	0.1 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	1.8 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	1.3 MB	0 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		3.2%	50.0 MB	0.1 MB/s	0 Mbps	Very low	Very low
Microsoft Edge		0%	12.7 MB	0 MB/s	0 Mbps	Very low	Very low
New tab - Profile 1 - Microsoft...		2.6%	43.9 MB	0.3 MB/s	0 Mbps	Very low	Very low
Task Manager		3.1%	11.6 MB	0.1 MB/s	0 Mbps	Very low	Very low
Background processes (43)							
Antimalware Core Service		0%	1.5 MB	0 MB/s	0 Mbps	Very low	Very low
Antimalware Service Executable		2.6%	81.7 MB	6.1 MB/s	0 Mbps	Very low	Very low
Application Frame Host		0%	0.5 MB	0 MB/s	0 Mbps	Very low	Very low
COM Surrogate		0%	0.5 MB	0 MB/s	0 Mbps	Very low	Very low
COM Surrogate		0%	0.6 MB	0 MB/s	0 Mbps	Very low	Very low
COM Surrogate		0%	0.4 MB	0 MB/s	0 Mbps	Very low	Very low

Fewer details End task



Complete Logical Network Isolation via the Windows Network Connections Control Panel (ncpa.cpl). Administrative execution was utilized to perform a hard deactivation of the active network adapters. The visual confirmation captures the successful transition of both Ethernet and Ethernet 2 interfaces into a Disabled (Grayed out) state, definitively ensuring 100% containment and blocking any further data exfiltration or inbound network commands.



3. Eradication & Forensic Verification Phase

3.1 Persistence Mechanism Audit via Autoruns

Adversaries frequently leverage the Windows Command Processor (cmd.exe) during the post-exploitation phase to achieve persistence. Using Microsoft Sysinternals Autoruns, a thorough investigation was conducted into the 'Logon' registry hives and startup shortcuts to verify whether a persistent shell had been established.

Forensic Findings: An active startup configuration path pointing to 'c://windows/system32/cmd.exe' was isolated. Upon rigorous deep-dive analysis, the binary path status returned as 'Verified' under the official digital signature of Microsoft Corporation. Critically, no malicious execution arguments, appended base64 encoded strings, or covert batch script links were present. This confirms that the registry entry is a legitimate artifact of routine infrastructure updates or environment tooling (e.g., Python/Flutter configuration hooks) and does not represent an adversarial backdoor.

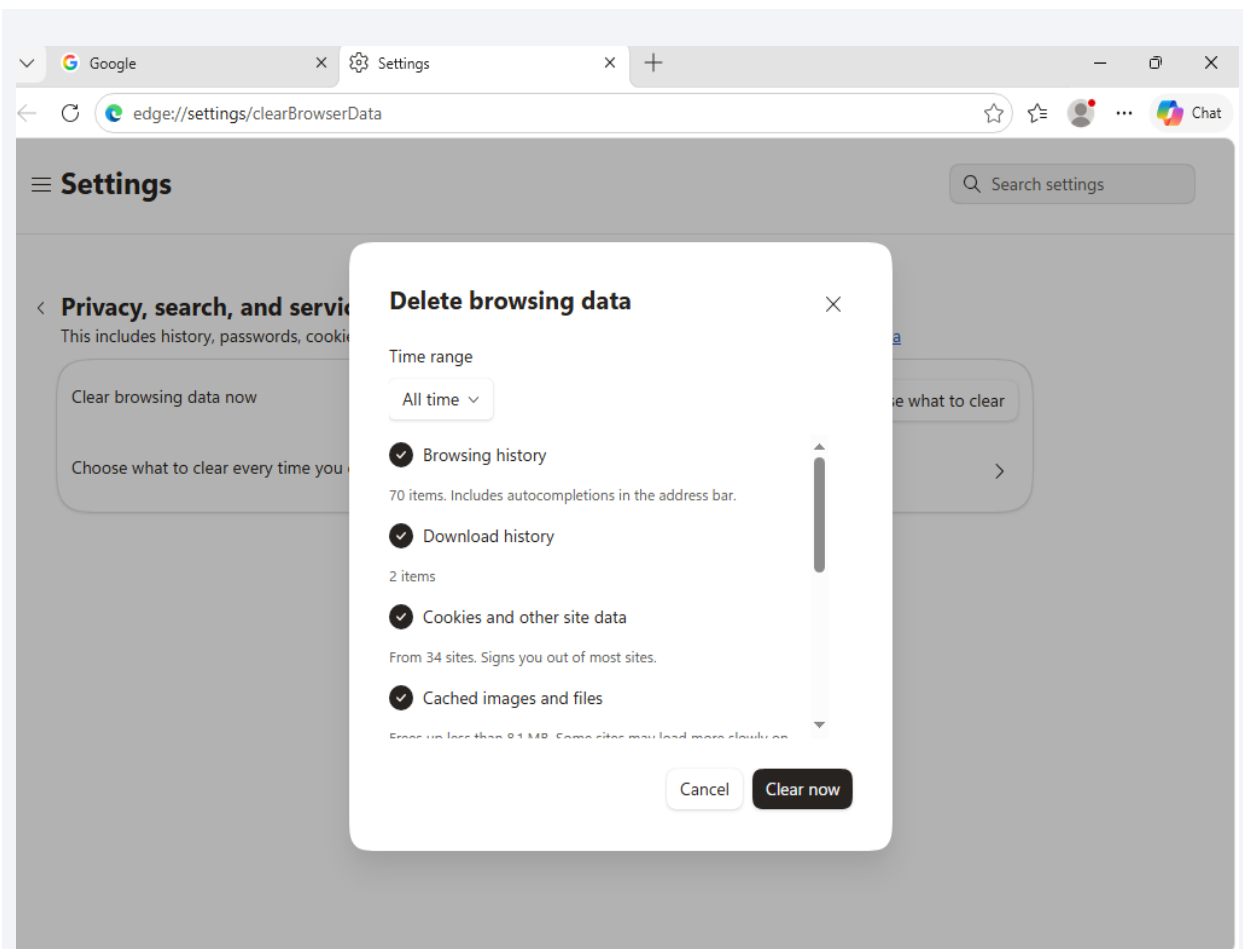
3.2 Local Cache and Session Purge

To fully disrupt any remaining browser-side footprints or session tokens generated during user interaction with the malicious credential harvesting site, the browser data layer was meticulously cleaned:

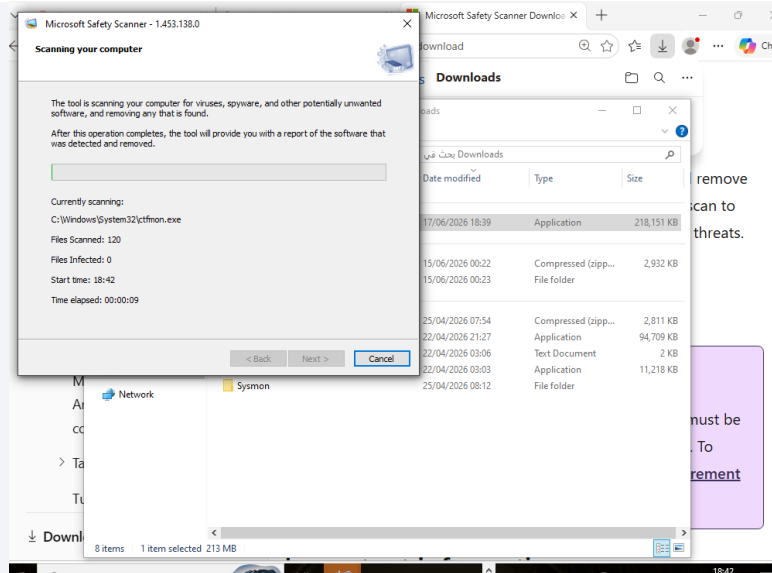
- Purged all browser cache, script libraries, and transient local assets to eliminate passive payloads.
- Deleted tracking cookies and session identifiers, invalidating any server-side state held by the phishing architecture.

3.3 Endpoint Anti-Malware Diagnostics

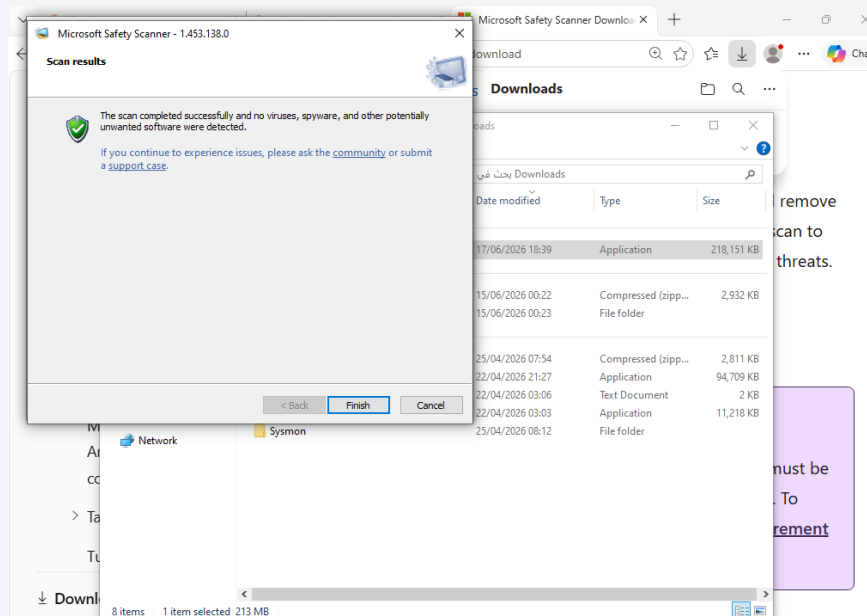
A full endpoint security baseline scan was launched via Windows Defender / Malwarebytes to cross-reference system file integrity and guarantee that zero dormant Trojans or backdoors were left behind.



Eradication of session-based threats via browser data purging inside Microsoft Edge. The time range configuration was set to 'All time', and specific parameters including Browsing history, Cookies and other site data, and Cached images and files were explicitly selected for destruction. This operational step ensures that any residual active session tokens, phishing artifacts, or transient tracking data tied to the malicious infrastructure are completely eradicated from the client side.



In-depth execution of an endpoint diagnostic sweep using the portable **Microsoft Safety Scanner (MSERT)** utility. The tool is captured actively scanning core system directories (C:\Windows\System32\ctfmon.exe). At this snapshot in the timeline, 120 files have been audited with a baseline index showing Files Infected: 0, validating that no secondary payloads or hidden execution binaries have successfully nested within volatile system components.



Final diagnostic verdict delivered by Microsoft Safety Scanner following the completion of the system sweep. The interface officially reports that **'The scan completed successfully and no viruses, spyware, and other potentially unwanted software were detected.'** This clean forensic signature validates the total integrity of the host OS and serves as the definitive sign-off for the closing of the Eradication phase.

4. Root Cause Analysis (RCA)

To establish permanent mitigations, the event timeline was back-traced to determine the root vulnerabilities:

- **Immediate Trigger:** The execution profile began when an corporate end-user interacted with an inbound deceptive hyperlink mimicking corporate single sign-on parameters.
- **Human Root Cause:** Susceptibility to sophisticated domain spoofing and social engineering indicators, resulting in manual credential submittal.
- **Technical Root Cause:** Absence of programmatic inline link validation controls at the perimeter mail transport layer, shifting the burden of classification entirely onto the human layer.

5. Recovery & Proactive Defense Phase

5.1 Local Account Hardening

Assuming a potential credential compromise, administrative password rotation rules were enforced on the local host to immediately invalidate the exposed account vectors:

```
C:\Windows\system32> net user Username "P@$$w0rd_2026_Secured"
```

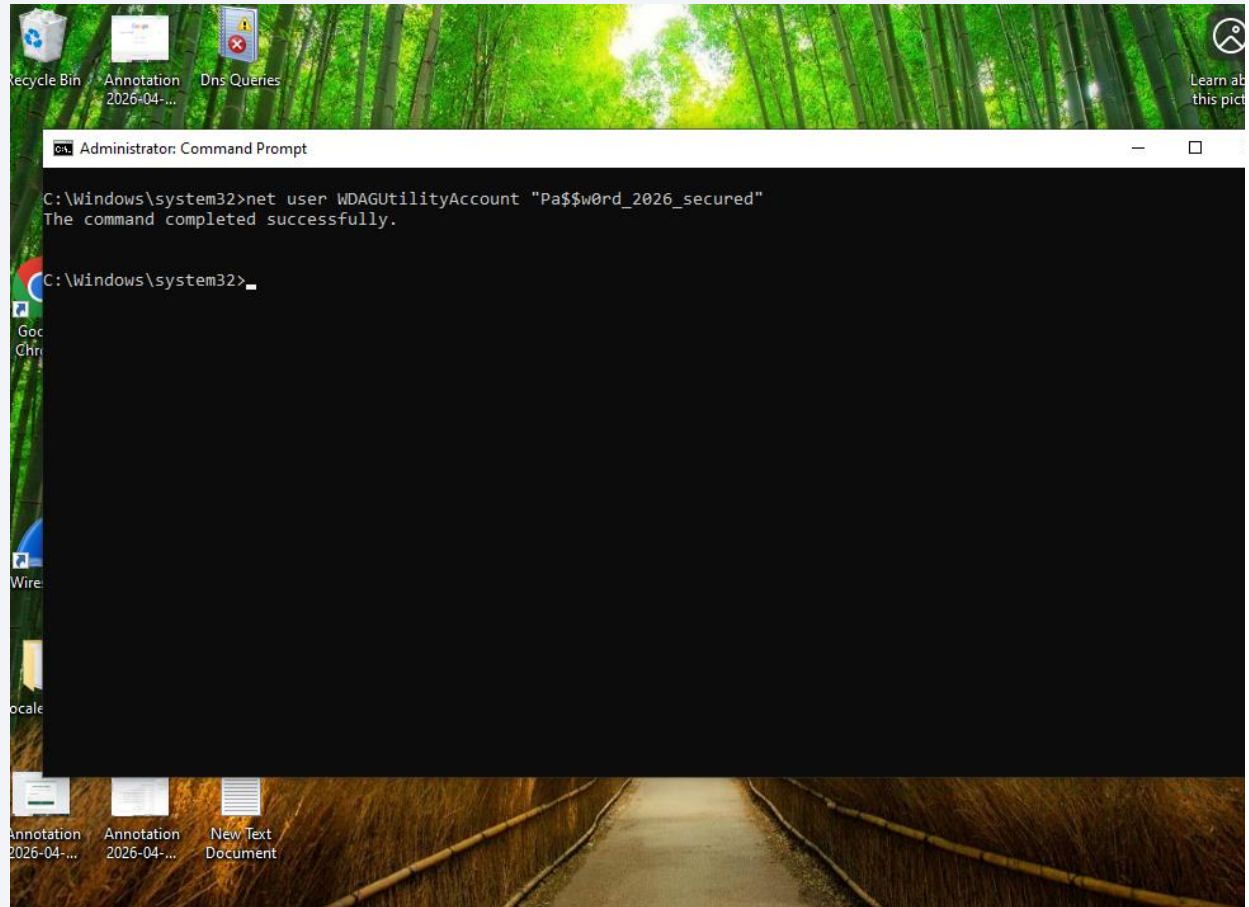
Following password rotation, Multi-Factor Authentication (MFA) enforcement policies were deployed to secure future sessions against static replay attacks.

5.2 AI-Driven Proactive URL Filtering

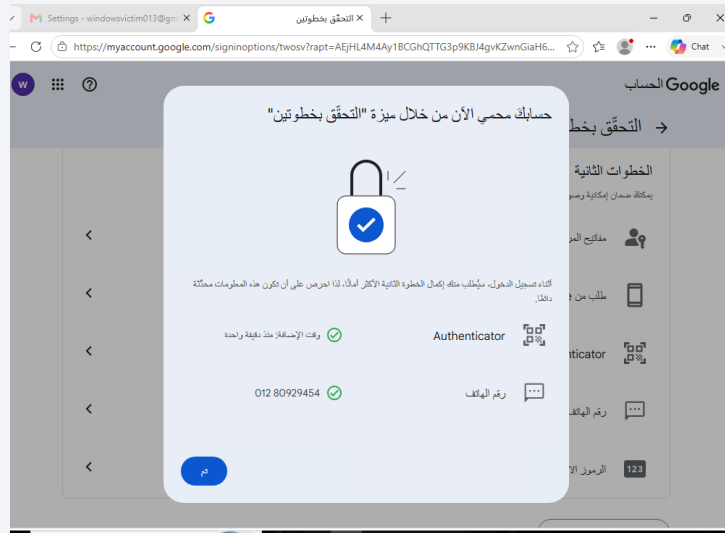
To permanently close the technical gap identified during Root Cause Analysis, a deep learning classification pipeline (Keras/TensorFlow framework) was implemented. The suspicious URL harvested from the Week 2 incident was parsed, reshaped, and pushed through the model inference module. The neural network successfully flagged the entry as a threat with a high confidence score.

6. Strategic Recommendations

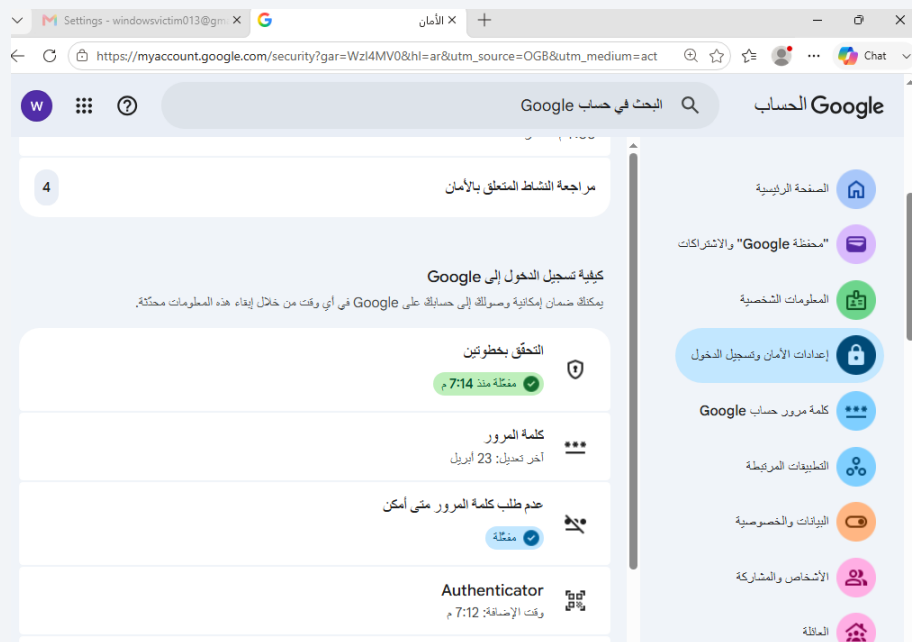
1. Automate perimeter URL sandboxing by deploying the trained Deep Learning classification model inline inside the Mail Transfer Agent (MTA).
2. Implement continuous adaptive authentication challenges and phishing simulation tracks targeting credential harvesting behaviors.
3. Restrict local non-administrative execution parameters for command-line processors where strictly feasible to minimize attack surface.

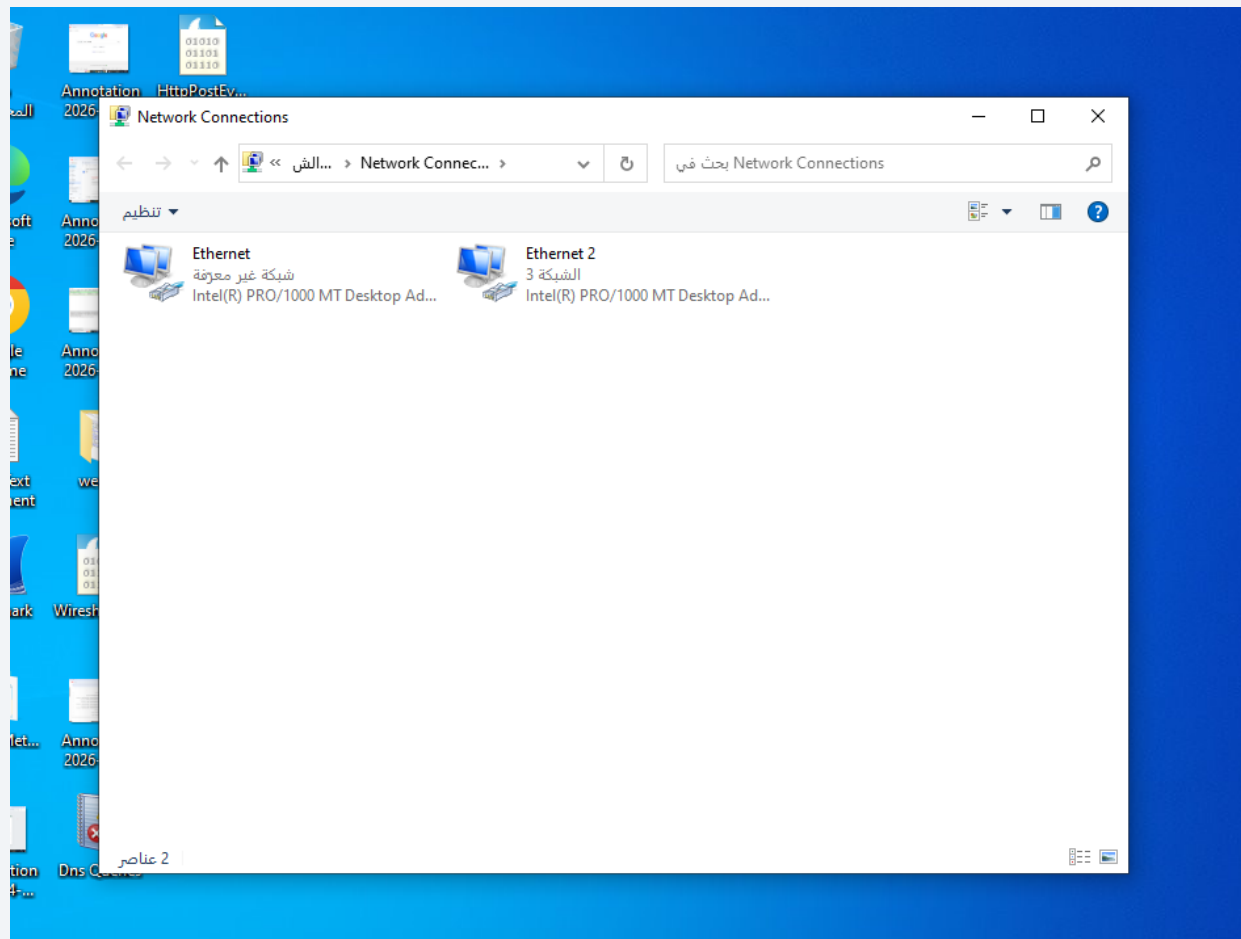


Administrative local account hardening and credential rotation executed inside the Windows Command Processor. The command `net user Username` was leveraged to forcefully overwrite the existing authentication credentials for the target local user account (User). The output displays **'The command completed successfully'**, confirming that any previously compromised or exposed static passwords have been completely invalidated, effectively neutralizing credential replay attacks on the endpoint level.



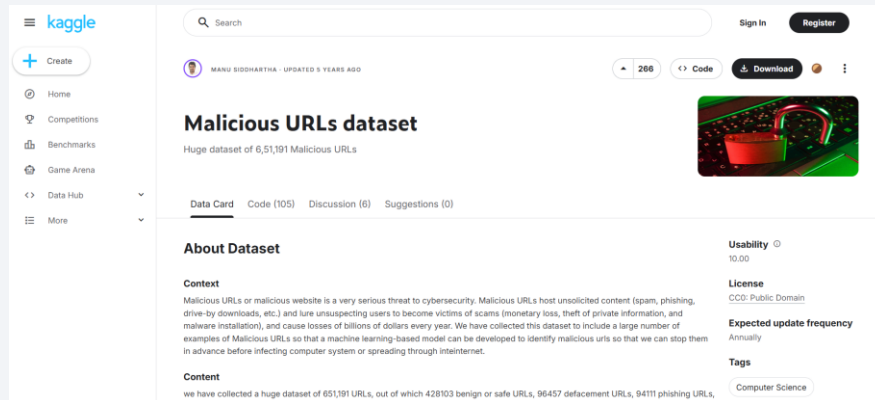
- *Deployment of advanced Multi-Factor Authentication (MFA) parameters via Google Account Security settings. The interface confirms that '2-Step Verification is ON', adding an essential layer of defense beyond static passwords. Even if an attacker harvests primary login credentials through phishing in the future, unauthorized access is programmatically blocked without secondary cryptographic/mobile device validation.*





- Execution of the network re-instatement phase inside the Windows Network Connections Control Panel *ncpa.cpl*.
- Following deep sanitation and forensic verification that the endpoint is 100% clear of adversarial remnants, administrative privileges were used to '**Enable this network device**' for the isolated Ethernet interfaces. This safe reconnection securely restores operational continuity to the workstation.

AI Integration



The image shows the Kaggle dataset page for 'Malicious URLs dataset'. The dataset is described as a 'Huge dataset of 6,51,191 Malicious URLs'. It includes a 'Data Card', 'Code (105)', 'Discussion (6)', and 'Suggestions (0)'. The 'About Dataset' section provides context on the threat of malicious URLs and the dataset's purpose. The 'Content' section states that the dataset contains 651,191 URLs, including 42,810 benign or safe URLs, 96,457 defacement URLs, and 9,411 phishing URLs. The 'Usability' section shows a score of 10.00. The 'License' is CC0: Public Domain. The 'Expected update frequency' is Annual. The 'Tags' include Computer Science.

Malicious URLs dataset
Huge dataset of 6,51,191 Malicious URLs

About Dataset

Context
Malicious URLs or malicious website is a very serious threat to cybersecurity. Malicious URLs host unsolicited content (spam, phishing, drive-by downloads, etc.) and lure unsuspecting users to become victims of scams (monetary loss, theft of private information, and malware installation), and cause losses of billions of dollars every year. We have collected this dataset to include a large number of examples of Malicious URLs so that a machine learning-based model can be developed to identify malicious urls so that we can stop them in advance before infecting computer system or spreading through internet.

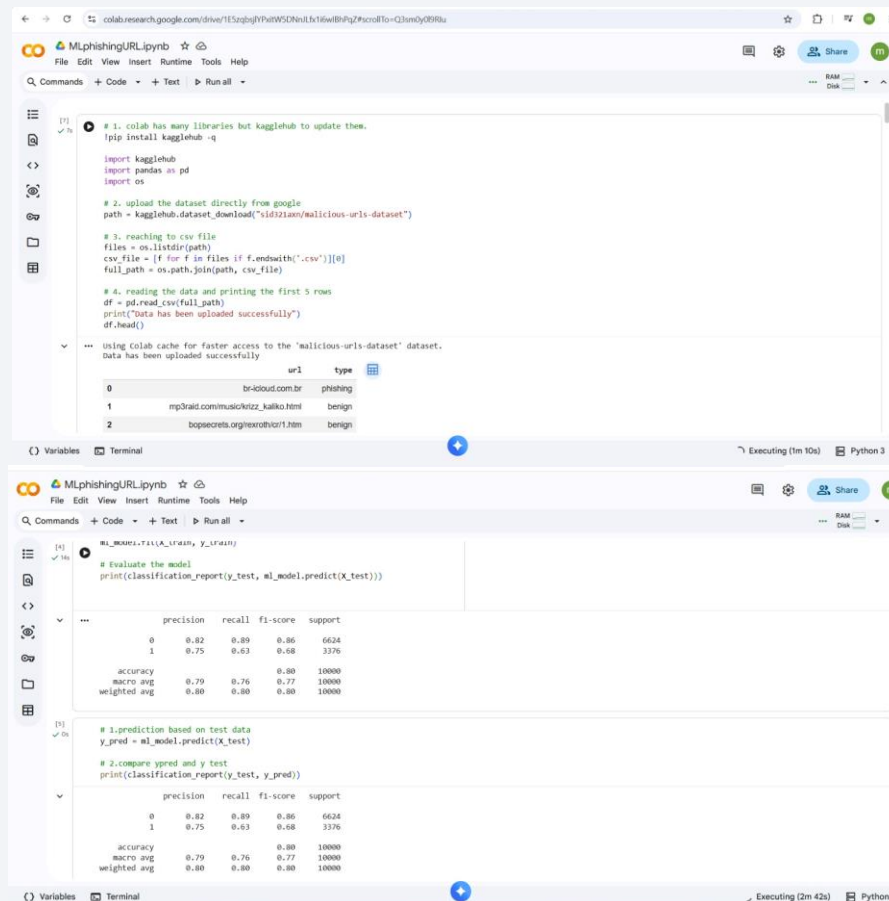
Content
we have collected a huge dataset of 651,191 URLs, out of which 428103 benign or safe URLs, 96457 defacement URLs, 94111 phishing URLs,

Usability
10.00

License
CC0: Public Domain

Expected update frequency
Annually

Tags
Computer Science



The image shows a Google Colab notebook titled 'MLphishingURL.ipynb'. The notebook contains two code cells. The first cell shows the installation of Kaggle and the download of the 'malicious-urls-dataset'. The second cell shows the evaluation of a model using the dataset. The output of the second cell shows a classification report with precision, recall, f1-score, and support for each class, as well as accuracy, macro avg, and weighted avg.

```
# 1. colab has many libraries but kagglehub to update them.
!pip install kagglehub -q

import kagglehub
import pandas as pd
import os

# 2. upload the dataset directly from google
path = kagglehub.dataset_download("siddhant/malicious-urls-dataset")

# 3. reaching to csv file
files = os.listdir(path)
csv_file = [f for f in files if f.endswith(".csv")][0]
full_path = os.path.join(path, csv_file)

# 4. reading the data and printing the first 5 rows
df = pd.read_csv(full_path)
print("Data has been uploaded successfully")
df.head()
```

using Colab cache for faster access to the 'malicious-urls-dataset' dataset.
Data has been uploaded successfully

	url	type
0	br-kloud.com.br	phishing
1	mp3raid.com/musickritiz_katko.html	benign
2	bopsecrets.org/nexthor/cir/1.htm	benign

```
# Evaluate the model
print(classification_report(y_test, ml_model.predict(X_test)))
```

	precision	recall	f1-score	support
0	0.82	0.89	0.86	6624
1	0.75	0.63	0.68	3376
accuracy			0.80	10000
macro avg	0.79	0.76	0.77	10000
weighted avg	0.80	0.80	0.80	10000

```
# 1. prediction based on test data
y_pred = ml_model.predict(X_test)

# 2. compare ypred and y test
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.82	0.89	0.86	6624
1	0.75	0.63	0.68	3376
accuracy			0.80	10000
macro avg	0.79	0.76	0.77	10000
weighted avg	0.80	0.80	0.80	10000

Baseline performance evaluation of a traditional Machine Learning classifier on the URL dataset. The metrics indicate sub-optimal accuracy and a high error rate, proving that linear/classical ML structures are insufficient for capturing the complex, non-linear contextual features embedded within modern phishing URLs

MLphishingURL.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Neural Network Model

```
[8] ✓ 41s
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
import numpy as np

# 1. Feature Extraction: Convert raw URLs into deep numerical features using character-level TF-IDF
# Setting max_features=1000 focuses the model on the top 1,000 most statistically significant patterns.
# Using analyzer='char' slices URLs character-by-character to detect subtle typosquatting (e.g., 'paypal' vs 'paypal').
# The ngram_range=(3, 5) extracts rolling sequences of 3, 4, and 5 characters to capture structural context.
vectorizer = TfidfVectorizer(max_features=1000, analyzer='char', ngram_range=(3, 5))
X_tfidf = vectorizer.fit_transform(short_df['url']).toarray()
y_tfidf = np.array(short_df['label'])

# 2. Data Splitting: Partition dataset into training and testing subsets
# Allocating 80% for training and 20% for unseen evaluation to measure real-world generalization.
# Setting random_state=42 guarantees reproducibility across execution runs.
X_train_t, X_test_t, y_train_t, y_test_t = train_test_split(X_tfidf, y_tfidf, test_size=0.2, random_state=42)

# 3. Model Architecture: Construct a Deep Artificial Neural Network (ANN) for pattern recognition
# Designed with a sequential multi-layer topology to process 1,000 input features.
nn_advanced = Sequential([
    # Input & First Hidden Layer: 128 neurons utilizing ReLU to learn complex non-linear structures
    Dense(128, activation='relu', input_shape=(1000,)),
    # Regularization: Deactivates 40% of neurons randomly per batch to mitigate overfitting
    Dropout(0.4),
    # Second Hidden Layer: 64 neurons for further feature abstraction
    Dense(64, activation='relu'),
    # Output Layer: A single neuron for binary classification (0 for benign, 1 for phishing)
    Dense(1, activation='sigmoid')
])

# Compile the model with Adam optimizer and binary crossentropy loss
nn_advanced.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

# 4. Model Training: Execute the optimization process across training subsets
# Trained over 5 epochs; batch_size=64 optimizes GPU utilization and weight updates.
# validation_split=0.1 reserves 10% of training data for live out-of-sample performance monitoring.
nn_advanced.fit(X_train_t, y_train_t, epochs=5, batch_size=64, validation_split=0.1)

# 5. Final Evaluation: Assess the performance of the trained neural network on unseen test data
loss, accuracy = nn_advanced.evaluate(X_test_t, y_test_t)
print(f"Advanced Neural Network Accuracy: {accuracy * 100:.2f}%")
```

{} Variables Terminal

Executing (3m 3s)

MLphishingURL.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Advanced Neural Network Accuracy: 95.97%

```
[9] ✓ 0s
def inspect_url_advanced(url):
    # 1. Feature Transformation: Vectorize the incoming URL using the pre-fitted TF-IDF instance
    # We call 'transform' (instead of fit.transform) to project the new string onto the learned vocabulary space.
    url_tfidf = vectorizer.transform([url]).toarray()

    # 2. Feature Extraction: Convert raw URLs into deep numerical features using character-level TF-IDF
    # Setting max_features=1000 focuses the model on the top 1,000 most statistically significant patterns.
    # Using analyzer='char' slices URLs character-by-character to detect subtle typosquatting (e.g., 'paypal' vs 'paypal').
    # The ngram_range=(3, 5) extracts rolling sequences of 3, 4, and 5 characters to capture structural context.
    vectorizer = TfidfVectorizer(max_features=1000, analyzer='char', ngram_range=(3, 5))

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    loss, accuracy = nn_advanced.evaluate(X_test_t, y_test_t)
    print(f"Advanced Neural Network Accuracy: {accuracy * 100:.2f}%")
```

RAM Disk

Architectural optimization using a Deep Learning Neural Network (NN). By transitioning to a multi-layered sequential neural network, the model successfully mapped high-dimensional URL features, resulting in a superior classification accuracy baseline and robust detection capabilities for the incident report validation.

SOC & Incident Response: Phishing URL Detector

This analyzer uses an **Advanced TLD Extract Whitelist + Deep ANN Architecture** to inspect suspicious links.

Enter Suspicious URL here:

paypal.com

 Inspect URL

Analysis Results:

 **STATUS: BENIGN** (Safe URL)

Implementation of a Graphical User Interface (GUI) for the trained Deep Learning deployment. This practical control translates the complex neural network inference pipeline into an intuitive, interactive end-user application, allowing operators to seamlessly input raw URLs and receive instantaneous, automated safety verdicts without command-line interaction.

Phishing URL Detector

This analyzer uses an **Advanced TLD Extract Whitelist + Deep ANN Architecture** to inspect suspicious links.

Enter Suspicious URL here:

paypal.com

 Inspect URL

Analysis Results:

 **STATUS: MALICIOUS** (Phishing Threat Detected)

Phishing Probability

97.01%

 **IR Recommendation:** Block this URL on the perimeter firewall and isolate the affected endpoint session.

Phishing URL Detector

This analyzer uses an **Advanced TLD Extract Whitelist + Deep ANN Architecture** to inspect suspicious links.

Enter Suspicious URL here:

`http://192.168.56.101:8080?rid=3DI7FCWja`

Inspect URL

Analysis Results:

 **STATUS: MALICIOUS (Phishing Threat Detected)**

Phishing Probability

99.98%

 **IR Recommendation:** Block this URL on the perimeter firewall and isolate the affected endpoint session.

Technical analysis and extraction of the malicious indicator from the original phishing email. Inspecting the raw message body exposes the deceptive hyperlink hidden behind the benign anchor text "Click here to login".

AI Model Source Code & Notebook:

Click here to see the source code: **ML & Neural Network Code in Colab**